# Criteria for detecting entanglement close to Dicke states with many-body correlations

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# Outline


1. Introduction

2. **Criteria with many-body correlations**
   - Bipartite criterion
   - Multiparticle entanglement

# Particle number resolving detection

- The resolution of the particle number detection is not 1 particle. Typically, $\sim 10$.
- So far we did not need single particle resolution.
- Particle-number resolving detection could improve the detected quality of the state dramatically.
- We could also have new entanglement criteria relying on single particle resolution.
- It is possible to reach a single-particle resolution:

M. Quensen, M. Hetzel, L. Santos, A. Smerzi, G. Tóth, L. Pezzé, C. Klempt, Hong-Ou-Mandel interference of more than 10 indistinguishable atoms, arXiv:2504.02691.

# Parity measurement

- We can measure the paritity as

$$\langle \sigma_z \otimes \sigma_z \otimes .... \otimes \sigma_z \rangle = \langle f(J_z) \rangle \,,$$

where

$$f(z) = e^{i2\pi(z+N/2)}.$$

- E. g, for $N = 4$, we have

$$\{f(z)\}_{z=-2,-1,0,1,2} = \{+1, -1, +1, -1, +1\}.$$

- Thus, we do not need individual access to the particles, but we need a particle number resolving detection.

# Entanglement conditions with many-body correlations

For separable states

$$\left|\langle \sigma_x \otimes \sigma_x \otimes \ldots \otimes \sigma_x \rangle\right| + \left|\langle \sigma_y \otimes \sigma_y \otimes \ldots \otimes \sigma_y \rangle\right| + \left|\langle \sigma_z \otimes \sigma_z \otimes \ldots \otimes \sigma_z \rangle\right| \leq 1$$

holds.

- For the ideal Dicke state the value is 3.

| $N$ | $\langle \sigma_x^{\otimes N} \rangle$ | $\lvert\langle \sigma_z^{\otimes N} \rangle\rvert$ | $\langle J_x^2 + J_y^2 \rangle$ | $\mathcal{J}$ | $(\Delta J_z)^2$ |
|---|---|---|---|---|---|
| 2 | 0.892(22) | 0.965(13) | 1.892(22) | 0.946(11) | 0.0176(66) |
| 4 | 0.821(44) | 0.951(25) | 5.08(29) | 0.85(5) | 0.025(12) |
| 6 | 0.833(61) | 0.942(33) | 11.26(85) | 0.94(7) | 0.029(17) |
| 8 | 0.821(70) | 0.806(70) | 19.0(16) | 0.95(8) | 0.098(36) |
| 10 | 0.872(72) | 0.822(86) | 25.7(26) | 0.86(9) | 0.091(45) |
| 12 | 0.61(13) | 0.862(96) | 33.7(46) | 0.80(11) | 0.067(44) |

**Extended Data Table 1**: Measurement results for various particle numbers. The uncertainties denote one standard deviation.

# Proof

For separable states

$$\left|\langle \sigma_x \otimes \sigma_x \otimes \ldots \otimes \sigma_x \rangle\right| + \left|\langle \sigma_y \otimes \sigma_y \otimes \ldots \otimes \sigma_y \rangle\right| + \left|\langle \sigma_z \otimes \sigma_z \otimes \ldots \otimes \sigma_z \rangle\right| \leq 1$$

holds.

- *Proof.* For a product state of the type

$$|\psi^{(1)}\rangle \otimes |\psi^{(2)}\rangle \otimes \ldots \otimes |\psi^{(N)}\rangle$$

  the left-hand side can be bounded from above as

$$\sum_{l=x,y,z} \left|\prod_{n=1}^{N} \left\langle \sigma_l^{(n)} \right\rangle\right| \leq \left|\left\langle \sigma_x^{(1)} \right\rangle \left\langle \sigma_x^{(2)} \right\rangle\right| + \left|\left\langle \sigma_y^{(1)} \right\rangle \left\langle \sigma_y^{(2)} \right\rangle\right| + \left|\left\langle \sigma_z^{(1)} \right\rangle \left\langle \sigma_z^{(2)} \right\rangle\right| \leq 1$$

  where in the first inequality we used that $\left|\left\langle \sigma_l^{(n)} \right\rangle\right| \leq 1$, and in the second inequality we used the Cauchy-Schwarz inequality and the fact that the length of the Bloch vector is at most one for a qubit.
- Separable states are mixtures of product states, hence the inequality is also valid for separable states. □

# States detected

- The witness also detects the GHZ states as entangled.
- It also detects the singlet state given as

$$\frac{1}{\sqrt{2}}(|01\rangle - |10\rangle) \otimes \frac{1}{\sqrt{2}}(|01\rangle - |10\rangle) \otimes ... \otimes \frac{1}{\sqrt{2}}(|01\rangle - |10\rangle)$$

  has

$$(\Delta J_z)^2 = 0,$$

  and

$$\left\langle \sigma_x^{\otimes N} \right\rangle = 1, \quad \left\langle \sigma_y^{\otimes N} \right\rangle = 1,$$

  if $N$ is divisible by 4. This is a 2-entangled state.
- Thus, these operators cannot be used to detect genuine multipartite entanglement.

# Inequality with multi-particle correlations

**Observation 1.** For $N$-qubit quantum states,

$$\langle J_x \rangle^2 / j^2 + \left\langle J_y \right\rangle^2 / j^2 + \left\langle \sigma_z^{\otimes N} \right\rangle^2 \leq 1$$

holds, where $j = N/2$ and

$$J_l = \frac{1}{2} \sum_{n=1}^{N} \sigma_l^{(n)}$$

for $l = x, y, z$.

*Proof.* The ground state of the Hamiltonian

$$H = BJ_x + K\sigma_z^{\otimes N},$$

where $B$ and $K$ are constants, is of the form

$$|\Psi\rangle = \alpha |0\rangle_x^{\otimes N} + \beta |1\rangle_x^{\otimes N},$$

which is a generalized GHZ state in the $x$-basis.

# Inequality with multi-particle correlations II

Then, the relevant expectation value of $J_x$ is

$$\langle J_x \rangle = \frac{N}{2} \langle \sigma_x \rangle_\phi$$

and the expectation value of the products of $\sigma_z$ matrices is

$$\left\langle \sigma_z^{\otimes N} \right\rangle = \langle \sigma_z \rangle_\phi \,,$$

where we define the single-qubit state

$$|\phi\rangle = \alpha |0\rangle_x + \beta |1\rangle_x .$$

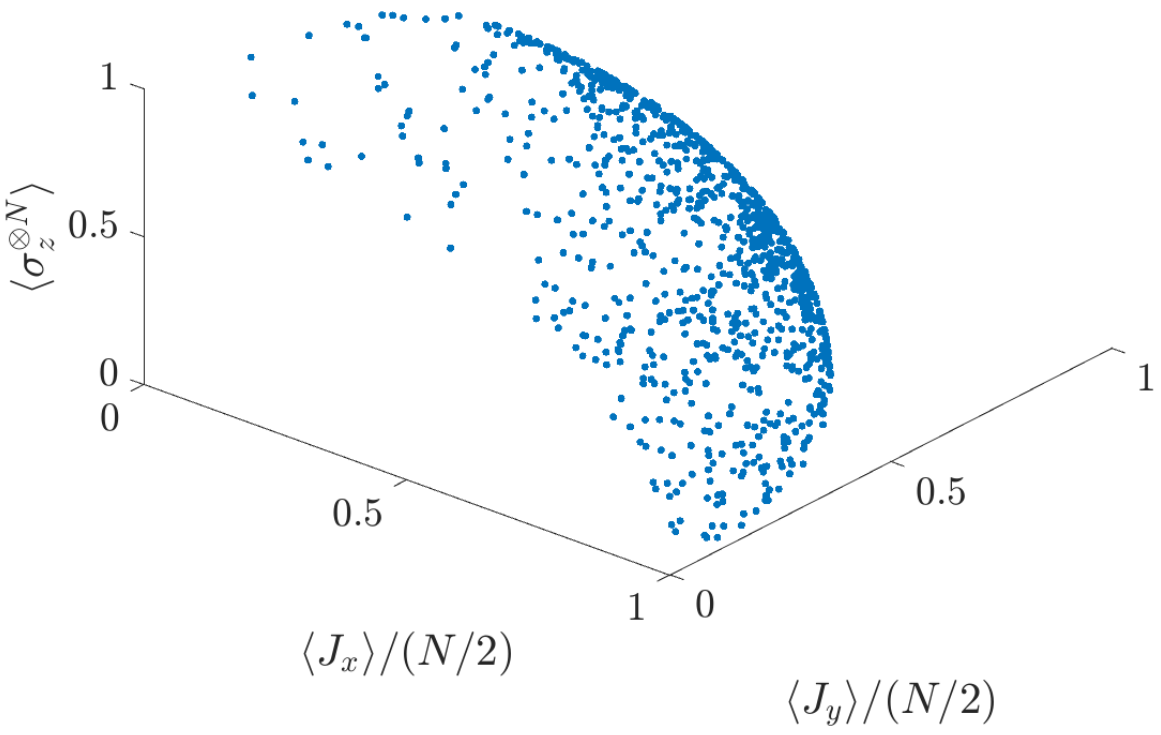
Since $\langle \sigma_x \rangle_\phi^2 + \langle \sigma_z \rangle_\phi^2 \leq 1$, it follows that

$$\langle J_x \rangle^2 / j^2 + \left\langle \sigma_z^{\otimes N} \right\rangle^2 \leq 1 .$$

Then, assuming that the mean spin is not parallel with the $x$-axis, but it is in the $xy$-plane, we arrive at our inequality. □

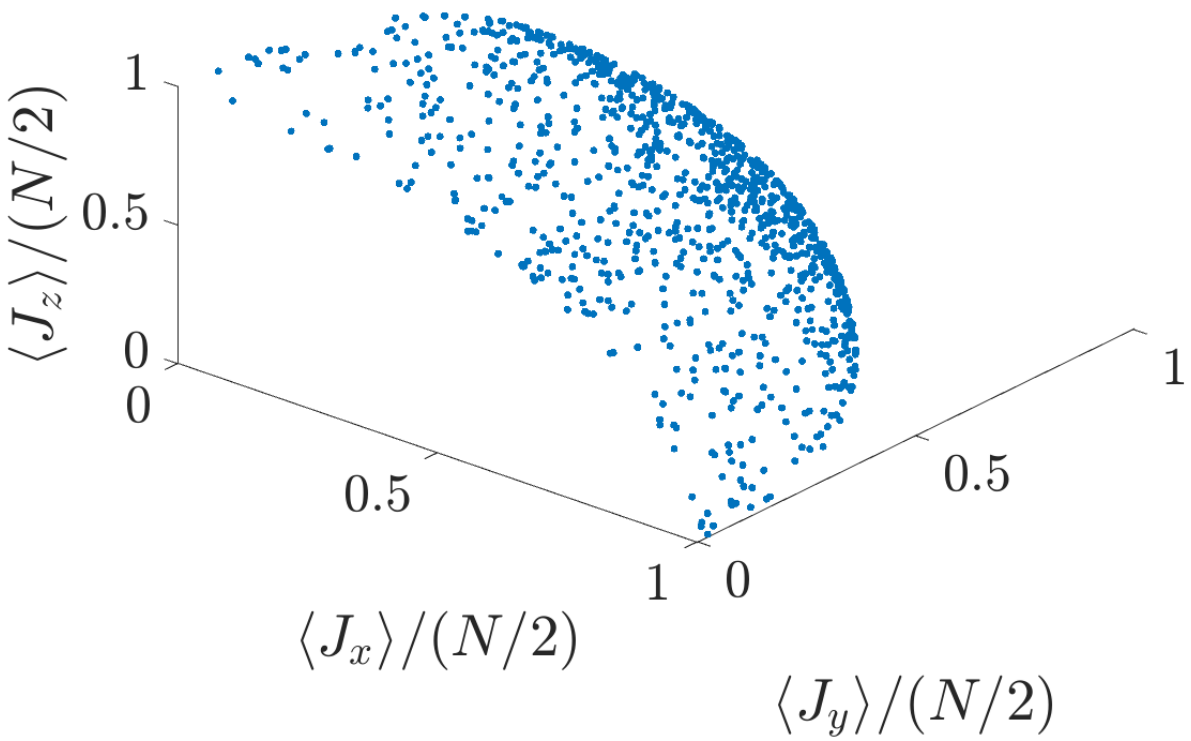
# Inequality with multi-particle correlations III

Generalized GHZ states:



# Inequality with multi-particle correlations IV

Comparison: spin coherent states



# Bipartite conditions

**Observation 2.** For bipartite separable states,

$$\langle J_x \otimes J_x \rangle / (j_1 j_2) + \left\langle J_y \otimes J_y \right\rangle / (j_1 j_2) + \left| \left\langle \sigma_z^{\otimes N_1} \otimes \sigma_z^{\otimes N_2} \right\rangle \right| \leq 1$$

holds, where for the left half we have

$$j_1 = N_1/2, \qquad j_2 = N_2/2.$$

| | |
|---|---|
| $N_1$ particles | $N_2$ particles |

*Proof.* We start from Observation 1

$$\langle J_x \rangle^2 / j^2 + \left\langle J_y \right\rangle^2 / j^2 + \left\langle \sigma_z^{\otimes N} \right\rangle^2 \leq 1$$

and use the Cauchy-Schwarz inequality. □

# Bipartite conditions

- Problem: we need to measure observables in the two halves of the system.
- In many experiments, we measure only collective observables.
- We need to modify the inequality such that it works for that case.
- Note that we need to measure the particle number with a single particle resolution.

# Bipartite conditions

**Observation 3.** The following expression is true for bipartite separable states

$$\left\langle J_x^2 + J_y^2 \right\rangle / (2j_1 j_2) + \left| \left\langle \sigma_z^{\otimes N} \right\rangle \right| \leq j(j+1)/(2j_1 j_2),$$

where

$$j_1 = N_1/2, \qquad j_2 = N_2/2, \qquad j = N/2.$$

*Proof.* We start from the previous Observation. We add to both sides

$$\left\langle (J_x^{(1)})^2 + (J_y^{(1)})^2 \right\rangle / (2j_1 j_2) + \left\langle (J_x^{(2)})^2 + (J_y^{(2)})^2 \right\rangle / (2j_1 j_2).$$

Then follows the relation

$$\begin{aligned} &\left\langle J_x^2 + J_y^2 \right\rangle / (2j_1 j_2) + \left| \left\langle \sigma_z^{\otimes N} \right\rangle \right| \\ &\leq 1 + \left\langle (J_x^{(1)})^2 + (J_y^{(1)})^2 \right\rangle / (2j_1 j_2) + \left\langle (J_x^{(2)})^2 + (J_y^{(2)})^2 \right\rangle / (2j_1 j_2). \end{aligned}$$

# Bipartite conditions II

Then, starting from the relation

$$\left\langle J_x^2 + J_y^2 \right\rangle / (2j_1 j_2) + \left| \left\langle \sigma_z^{\otimes N} \right\rangle \right| \\ \leq 1 + \left\langle (J_x^{(1)})^2 + (J_y^{(1)})^2 \right\rangle / (2j_1 j_2) + \left\langle (J_x^{(2)})^2 + (J_y^{(2)})^2 \right\rangle / (2j_1 j_2),$$

we use the inequality

$$\left\langle (J_x^{(n)})^2 + (J_y^{(n)})^2 \right\rangle \leq j_n(j_n + 1).$$

We arrive at

$$\left\langle J_x^2 + J_y^2 \right\rangle / (2j_1 j_2) + \left| \left\langle \sigma_z^{\otimes N} \right\rangle \right| \leq j(j+1)/(2j_1 j_2).$$

We need to measure only collective quantities! □

# Outline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2. **Criteria with many-body correlations**
   - Bipartite criterion
   - Multiparticle entanglement

# Conditions for multi-particle entanglement

**Observation 4.** States violating the inequality

$$\left\langle J_x^2 + J_y^2 \right\rangle / (2j_1 j_2) + \left| \left\langle \sigma_z^{\otimes N} \right\rangle \right| \leq j(j+1)/(2j_1 j_2),$$

for

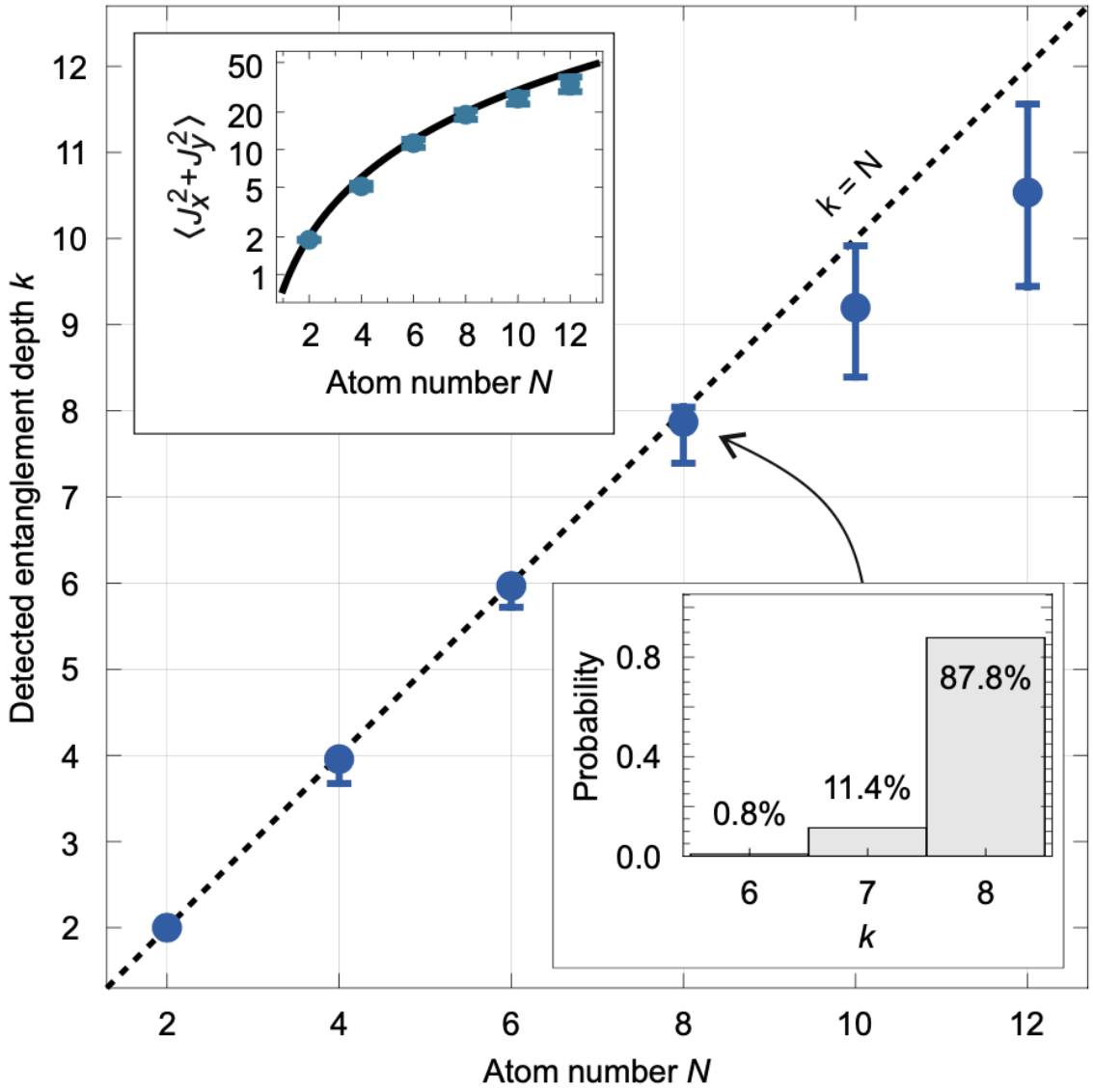
$$j_1 = k/2, \qquad j_2 = (N-k)/2$$

| | |
|---|---|
| $k$ particles | $N-k$ particles |

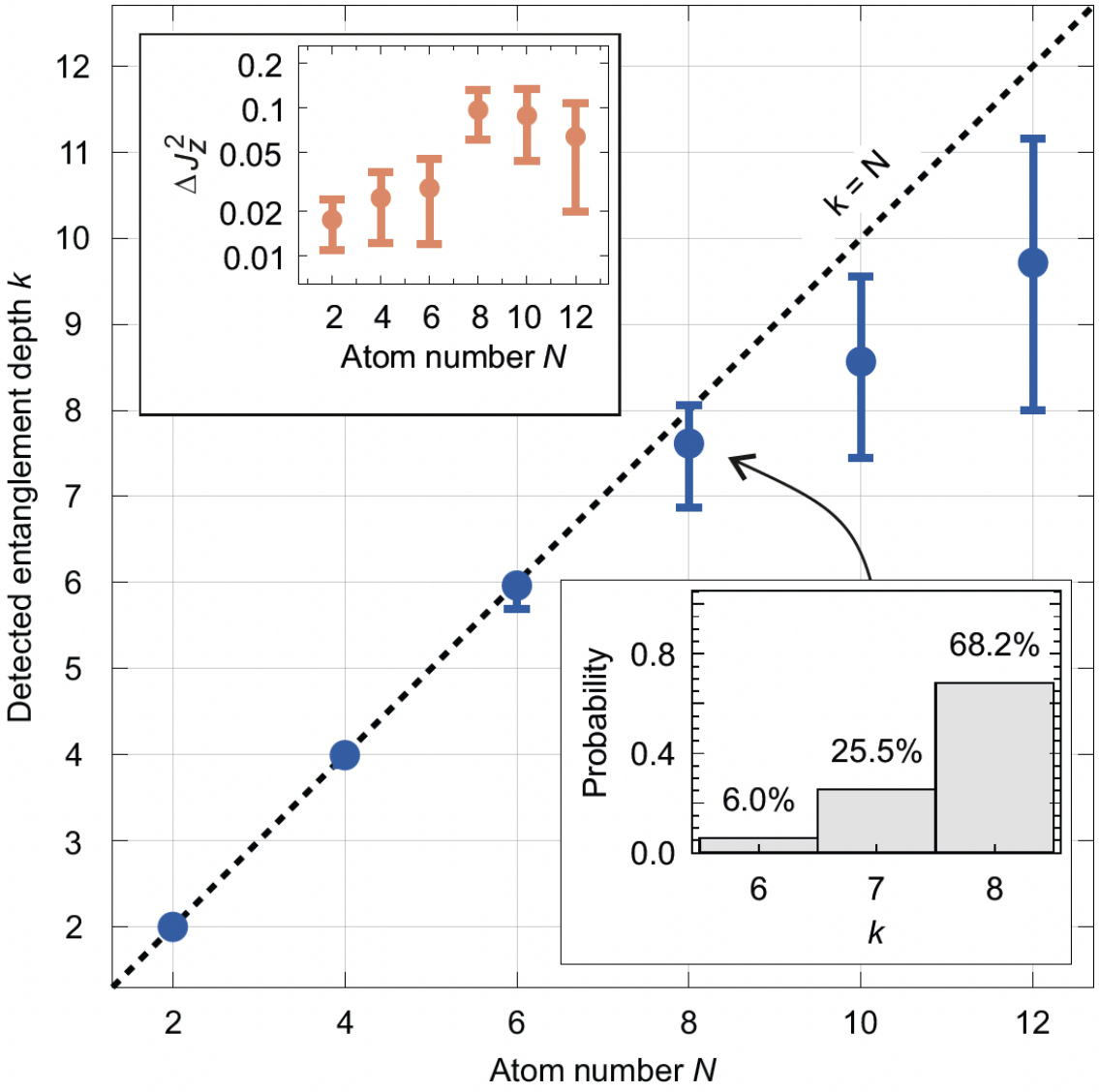
possess at least $(k+1)$-particle entanglement, where we assume that $k \geq N/2$.

Violation for $k = N-1$ means genuine multipartite entanglement.

# Results


Detected entanglement depth k
Atom number N
k = N
⟨J_x^2+J_y^2⟩
Atom number N
Probability
0.8%
11.4%
87.8%
k

# Comparison to the alternative method



# Conclusions

- We discussed how to detect bipartite and multipartite entanglement with many-body correlation measurements.
- The method has been successfully used in experiments with Dicke states up to 12 particles.
- It demonstrates the good quality of the created Dicke state.
- For the transparencies, see

www.gtoth.eu

- See also

M. Quensen, M. Hetzel, L. Santos,A. Smerzi,
G. Tóth, L. Pezzé, C. Klempt.

Hong-Ou-Mandel interference of more than 10 indistinguishable atoms,

arXiv:2504.02691.

THANK YOU FOR YOUR ATTENTION!